MTH219 2021 S2 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ICT133 Structured Programming
* ICT162 Structured Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

Donaldson Tan

Answer all questions. (Total 100 marks)

Question 1

(a) x_i for $i = 1, \ldots, 51$ are the number of apples sold in a fruit store in 51 different days. You may assume that the number of apples sold for each day is independent from one another.

$$\sum_{i=1}^{51}(x_i - 70) = 676$$

$$\sum_{i=1}^{51}(x_i - 70)^2 = 23718$$

(i) Calculate the mean and variance of the dataset.

(5 marks)

(ii) Using the value of the variance calculated in Question 1(a)(i), calculate the value of $\sum_{i=1}^{51} x_i^2$.

(4 marks)

(b) Cindy throws a fair cubic die with faces numbered 1, 2, 3, 4, 5, 6. If she throws an odd number then her score is the number thrown. If she throws an even number, she will throw the die a second time and her score is the sum of both numbers drawn (Figure Q1(b)).

Let the random variable X denote Cindy's score.

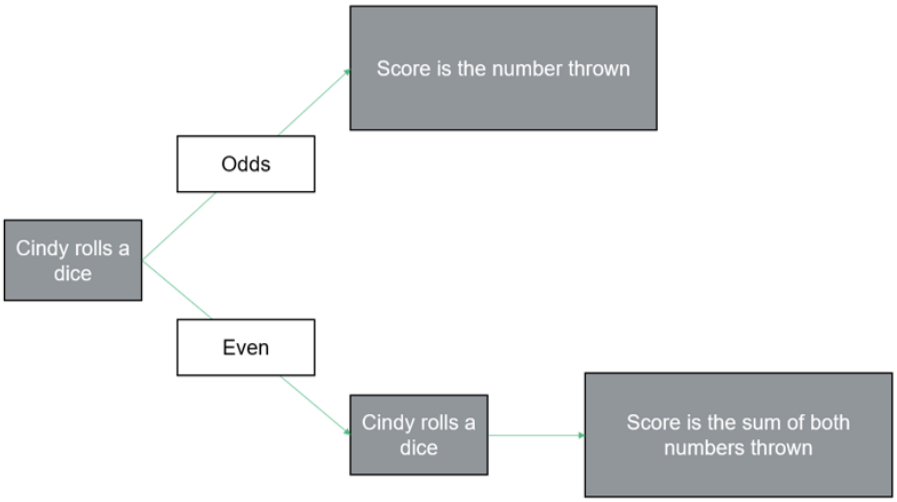


Figure Q1(b)

(i) Compute the frequency table for Cindy's score. You may find it helpful to create a table with all possible dice rolls and the scores.

x	1	2	3	4	5	6	7	8	9	10	11	12
Frequency, f												
Probability, p(x)												

(8 marks)

(ii) And hence computer $E(X)$ and $Var(X)$.

Hint:

$$E(X) = \sum x\, P(x) \;\; for\ all\ X$$

$$E(X^2) = \sum x^2\, P(x)\, for\ all\ X$$

(8 marks)

Question 2

Bag A contains 4 balls numbered 1, 3, 5, 7.
Bag B contains 5 balls numbered 5, 7, 7, 7, 9.
Bag C contains 7 balls numbered 1, 4, 6, 7, 7, 7, 7.

One ball is selected at random from each bag.

Event X is 'exactly two of the selected balls have the same number'.
Event Y is 'the ball selected from bag C has number 6'.

(a) (i) Compute the probability that 2 "Ones" will be drawn.

(2 marks)

(ii) Compute the probability that 2 "Fives" will be drawn.

(2 marks)

(iii) Compute the probability that 2 "Sevens" will be drawn.

(4 marks)

(iv) Hence or otherwise, compute $P(X)$.

(4 marks)

(b) Describe in words the event $X \cap Y$. What are all the possible scenarios?

(3 marks)

(c) Compute $P(X \cap Y)$ and hence determine whether events X and Y are independent.

(6 marks)

(d) Compute the probability that two balls are numbered 5, given that exactly two of the selected balls have the same number.

(4 marks)

Question 3

A business owner set up an online shop with an e-Commerce Site. Based on his previous observations, he found that items were purchased at the following rates:

Day/Night	Time	Rate
Day	9 am to 9 pm	20 per hour
Night	9 pm to 9 am	5 per hour

Table Q3

(a) It is proposed that instead of splitting the day into Day and Night, we can simplify the analysis by considering the purchase rate on a daily basis instead.

(i) What is the daily average rate? (Ignore time)

(2 marks)

(ii) Explain if the Poisson Distribution is an appropriate model to study the purchases over the entire day (24 hours, ignoring time).

(5 marks)

(b) Let X be the number of purchases in Day time (9 am to 9 pm). Poisson distribution may be assumed to be suitable model for X.

(i) Apply the Poisson Distribution and compute the probability for exactly 8 purchases in any half hour.

(3 marks)

(ii) Let T be the waiting time to the next purchase in the Day time (9 am to 9 pm). Write down the distribution of T.

(2 marks)

(iii) What is the expected value of T?

(2 marks)

(iv) Using the CDF of T or otherwise, calculate $P(T > 5)$, the probability that the waiting time to the next purchase exceeds 5 minutes.

(4 marks)

(c) We will now model the number of purchases at the night time. There are an average of 5 purchases per hour from 9 pm to 9 am. Assume that the Poisson distribution is a suitable model.

From 9 pm to 9 am, there are 12 "1 hour" periods, specifically:

1. 9 pm to 10 pm;
2. 10 pm to 11 pm;
   …;
12. 8 am – 9 am

(i) What is the probability for a single "one hour" period to have at least 5 purchases?

(4 marks)

(ii) What is the probability that for all "1 hour" periods to have at least 5 purchases each?

(3 marks)

Question 4

(a) The probability density function of the random variable X is given by

$$f(x) = \begin{cases} \frac{2}{9}\, x\,(3-x) & 0 \le x \le 3, \\ 0 & Otherwise. \end{cases}$$

(i) Sketch the graph of $y = f(x)$ and state the median of X.

(4 marks)

(ii) A statistical analyst wishes to compute the probability of $X < 1.5$. With the help of the graph drawn in Question 4(a)(i), show how can he compute $P(X < 1.5)$.

(2 marks)

(iii) Given that $P(X > 0.5) = 0.9259$, compute the value of $P(0.5 < X < 1.5)$.

(2 marks)

(b) The time Caden spends on gaming each day has a normal distribution with mean 1.8 hours and standard deviation σ hours. On 90% of these days, he spends more than 1.3 hours on his gaming.

(i) Determine the value of σ.

(3 marks)

(ii) Compute the probability that, on a randomly chosen day, Caden spends less than 2 hours on gaming.

(3 marks)

(c) The MTH219 module assessment weighting in 2020 was

Assessment	Weight Allocation	Marks	
		Mean	Standard Deviation
Examination	70%	53.2	9.3
CMA	8%	85.6	15.7
TMA	16%	65.4	19.6
PCQ	6%	95.3	8.3

Table Q4(c)

(i) Calculate the mean and standard deviation of the overall marks (weighted average of the four assessments). Assume that the marks in each assessment are independent from one another.

(5 marks)

(ii) What is the proportion of students who are expected to score an A grade, if A grade is 75 marks and above?

(3 marks)

(iii) Do you think that the assumption "the scores from the 4 assessments are independent" is reasonable? State your reasoning clearly.

How will the mean and standard deviation of the overall marks be affected if the marks in each assessment are not independent from one another?

(3 marks)

----- END OF PAPER -----

1(a) i. $\sum_{i=1}^{51}(x_i-70) = 676$

$\sum_{i=1}^{51} x_i - 51(70) = 676$

$\sum_{i=1}^{51} x_i = 676 + 51(70) = 4246$

$\therefore \bar{x} = \frac{1}{51}\sum_{i=1}^{51} x_i = \frac{4246}{51} \simeq 83.2549$

$\sum_{i=1}^{51}(x_i-70)^2 = 23718$

$$\begin{aligned}
Var(X-70) &= E([X-70]^2) - [E(X-70)]^2 \\
&= \frac{1}{51}\sum_{i=1}^{51}(x_i-70)^2 - \left[\frac{1}{51}\sum_{i=1}^{51}(x-70)\right]^2 \\
&= \frac{23718}{51} - \left(\frac{676}{51}\right)^2 \\
&= 289.3664
\end{aligned}$$

$Var(X) = Var(X-70) = 289.3664$

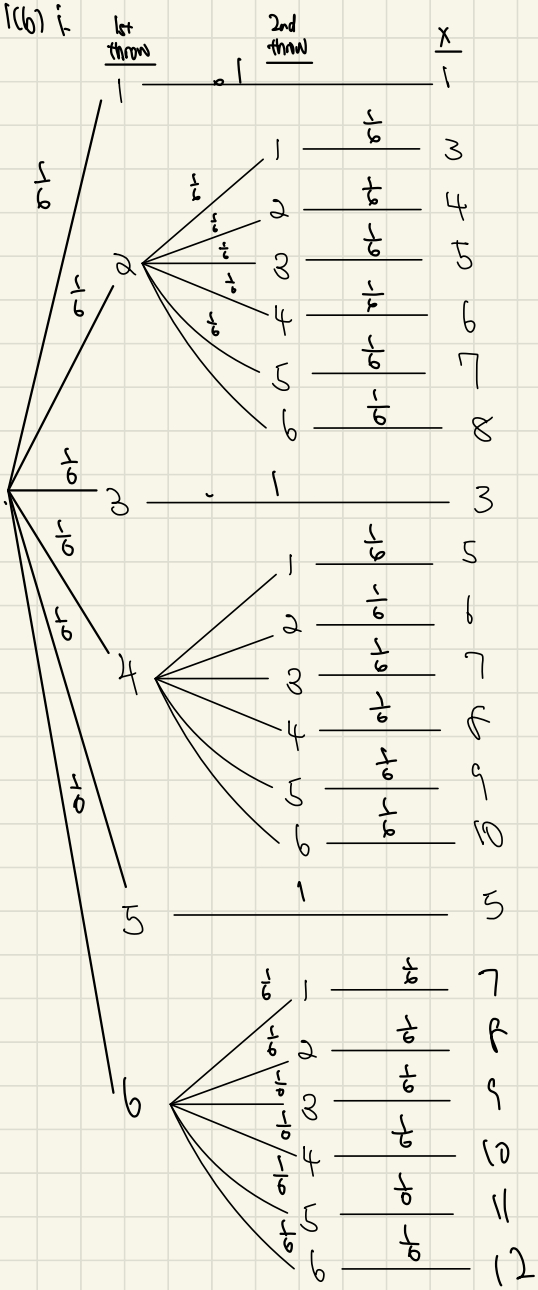
ii. $Var(X) = E(X^2) - [E(X)]^2$

$E(X^2) = Var(X) + [E(X)]^2$

$\frac{1}{51}\sum_{i=1}^{51} x_i^2 = 289.3664 + \left(\frac{4246}{51}\right)^2 = 7220.745$

$\sum_{i=1}^{51} x_i^2 = 51 \times 7220.745 = 368258$

1(b) i.



$P(X=1) = \frac{1}{6} = \frac{6}{36}$

$P(X=2) = 0$

$P(X=3) = \frac{1}{6}\cdot\frac{1}{6} + \frac{1}{6} = \frac{7}{36}$

$P(X=4) = \frac{1}{6}\cdot\frac{1}{6} = \frac{1}{36}$

$P(X=5) = 2\times\frac{1}{6}\cdot\frac{1}{6} + \frac{1}{6} = \frac{8}{36}$

$P(X=6) = 2\times\frac{1}{6}\cdot\frac{1}{6} = \frac{2}{36}$

$P(X=7) = 3\times\frac{1}{6}\cdot\frac{1}{6} = \frac{3}{36}$

$P(X=8) = 3\times\frac{1}{6}\cdot\frac{1}{6} = \frac{3}{36}$

$P(X=9) = 2\times\frac{1}{6}\cdot\frac{1}{6} = \frac{2}{36}$

$P(X=10) = 2\times\frac{1}{6}\cdot\frac{1}{6} = \frac{2}{36}$

$P(X=11) = \frac{1}{6}\cdot\frac{1}{6} = \frac{1}{36}$

$P(X=12) = \frac{1}{6}\cdot\frac{1}{6} = \frac{1}{36}$

1(b) i

x	1	2	3	4	5	6	7	8	9	10	11	12
Frequency, f	6	0	7	1	8	2	3	3	2	2	1	1
Probability, p(x)	$\frac{6}{36}$	0	$\frac{7}{36}$	$\frac{1}{36}$	$\frac{8}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{2}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

(8 marks)

ii. $E(X) = \sum_{k=1}^{12} k.P(X=k) = 5.25$

$E(X^2) = \sum_{k=1}^{12} k^2 P(X=k) = 36.75$

$Var(X) = E(X^2) - [E(X)]^2$

$= 36.75 - 5.25^2$

$= 9.1875$

2. Bag A contains 4 balls numbered 1, 3, 5, 7.
Bag B contains 5 balls numbered 5, 7, 7, 7, 9.
Bag C contains 7 balls numbered 1, 4, 6, 7, 7, 7, 7.

(a) total no. of ways to pick 3 balls $= 4 \times 5 \times 7 = 140$

no. of ways to pick 2 '1's $= 1 \times 1 \times 5 = 5$

no. of ways to pick 2 '5's $= 1 \times 1 \times 7 = 7$

no. of ways to pick 2 '7's $= 1 \times 3 \times 3 + 3 \times 3 \times 4 + 1 \times 1 \times 4$
$= 53$

$P(\text{pick 2 ones}) = \frac{5}{140}$

$P(\text{pick 2 fives}) = \frac{7}{140}$

$P(\text{pick 2 'sevens'}) = \frac{53}{140}$

$P(X) = P(\text{pick 2 ones}) + P(\text{pick 2 fives}) + P(\text{pick 2 sevens})$
$= \frac{5}{140} + \frac{7}{140} + \frac{53}{140}$
$= \frac{65}{140}$

2(b) 6 is found in C but not in A & B. Hence $X \cap Y$ describes outcome that the same number was picked from A, B while 6 was picked from C.

$\therefore$ sample space $= \{(5,5,6), (7,7,6)\}$

(c) no. of ways to pick $(5,5,6) = 1 \times 1 \times 1 = 1$

no. of ways to pick $(7,7,6) = 1 \times 3 \times 1 = 3$

$$P(X \cap Y) = \frac{1+3}{140} = \frac{4}{140} = \frac{1}{45}$$

no. of ways to pick $6 = 4 \times 5 \times 1 = 20$

$$P(Y) = \frac{20}{140} = \frac{1}{7}$$

$$P(X) \cdot P(Y) = \frac{11}{20} \cdot \frac{1}{7} = \frac{11}{140}$$

$$P(X \cap Y) \neq P(X) \cdot P(Y)$$

Hence, events X and Y are not independent

(d) $P(2 \text{ fives} \mid 2 \text{ of the selected balls have same number})$

$$= \frac{7}{140} \div \frac{65}{140}$$

$$= \frac{7}{65}$$

3(a) i. daily purchase rate $= 12(20) + 12(5)$

$= 300$ per day

ii. Poisson is not an appropriate model because the daily purchase rate is not scalable when the time interval is reduced from 24h to 12h. In fact, the purchase rate is 20/h from 9am to 9pm and 5/h from 9pm to 9am.

(b) $X \sim Po(10)$

i $P(X=8) = \text{dpois}(8, 10)$

$= 0.112599$

ii. $T \sim Exp(\lambda)$ where $\lambda = 20/h$

iii $E(T) = \frac{1}{\lambda} = \frac{1}{20}$ h

iv. $P(T > 5\text{min}) = P(T > \frac{1}{12}h)$

$= 1 - P(T \leq \frac{1}{12}h)$

$= 1 - \text{pexp}(\frac{1}{12}, 20)$

$= 0.1888756$

3(c) let Y be the no. of purchase during 1h interval of night time.

$$Y \sim Po(5)$$

i. $P(Y \geq 5) = 1 - P(Y \leq 4)$

$= 1 - \text{ppois}(4, 5)$

$= 0.5595067$

ii. let $p = P(Y \geq 5) = 0.5595067$

let X be the no. of 1h intervals during night time with at least 5 purchase,

$$X \sim Bin(12, p)$$

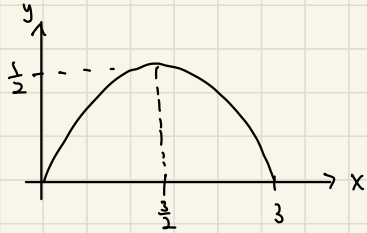
P(all 12 intervals have at least 5 purchases)

$= P(X = 12)$

$= \text{dbinom}(12, 12, p)$

$= 0.000941160 4$

4(a) i.



$$y = f(x) = \frac{2}{9}x(x-3)$$

Since quadratic equation is symmetric, then the median must be at the line of symmetry

$$\therefore \text{median} = \frac{0+3}{2} = \frac{3}{2}$$

$$f(\text{median}) = \frac{2}{9}\left(\frac{3}{2}\right)\left(\frac{3}{2}\right) = \frac{1}{2}$$

ii. Since $x = \frac{3}{2}$ is the median, then $P(X < \frac{3}{2}) = 0.50$

iii. $P(0.5 < X < 1.5) = P(X < 1.5) - P(X < 0.5)$

$$= P(X < 1.5) - (1 - P(X > 0.5))$$

$$= 0.4259$$

(b) i. let X be time spent on gaming

$$X \sim N(1.8, \sigma^2)$$

$$P(X > 1.3) = 0.90 \Rightarrow P(X \leq 1.3) = 0.10$$

$$\frac{1.3 - 1.8}{\sigma} = \text{qnorm}(0.10)$$

$$\therefore \sigma = \frac{1.3 - 1.8}{\text{qnorm}(0.10)} = 0.3901521$$

ii. $P(X < 2) = \text{pnorm}(2, 1.8, \sigma)$

$$= 0.6958916$$

4(c) i.

$$\text{overall} = 0.70 \times \text{examination} + 0.08 \times \text{CMA} + 0.16 \times \text{TMA} + 0.06 \times \text{PCQ}$$

$$E(\text{overall}) = 0.70 \cdot E(\text{examination}) + 0.08 \cdot E(\text{CMA}) + 0.16\, E(\text{TMA}) + 0.06\, E(\text{PCQ})$$

$$= 60.27$$

$$\text{var}(\text{overall}) = \text{var}(0.70 \times \text{examination}) + \text{var}(0.08 \times \text{CMA}) + \text{var}(0.16 \times \text{TMA}) + \text{var}(0.06 \times \text{PCQ})$$

$$= 0.70^2 \times \text{var}(\text{examination}) + 0.08^2 \times \text{var}(\text{CMA}) + 0.16^2 \times \text{var}(\text{TMA}) + 0.06^2 \times \text{var}(\text{PCQ})$$

$$= 54.04014$$

$$\text{standard deviation} = \sqrt{\text{var}(\text{overall})}$$

$$= 7.3512$$

4(c) ii.

$$\text{overall} \sim N(60.27, 54.04014)$$

$$\begin{aligned} P(\text{overall} \geq 75) &= 1 - P(\text{overall} < 75) \\ &= 1 - \text{pnorm}(75, 60.27, 7.3512) \\ &= 0.02254818 \\ &\simeq 0.0225 \end{aligned}$$

$\therefore$ 0.0225 of the students are expected to score A

iii. The assumption that each same component is independent is not valid. This is because the stronger students will perform well for each assessable component, while the weaker students will perform poorly for each assessable component.